Unit 5 Quiz: Study Design KEY

Name _____

Show your work.

5.1 - Samples, Studies, and Bias

1. **I name the population and the sample in a study, and I tell a sample survey, an observational study, and an experiment apart.**

   A music app has 60,000 users. To learn how users feel about a new design, the app sends a survey to 500 randomly chosen users.

   (a) What is the population?

   All 60,000 users of the app

   (b) What is the sample?

   The 500 users who got the survey

   Is each study a sample survey, an observational study, or an experiment? Explain.

Study	Type of study	Explain
(c) The music app study above	Sample survey	It asks a sample questions. It changes nothing.
(d) A researcher counts the birds that visit 20 yards with trees and 20 yards with no trees.	Observational study	She only watches and counts. She does not change any trees.
(e) A teacher randomly chooses half of her students to read a story on paper. The others read it on a screen. All take the same quiz.	Experiment	She assigns a treatment (paper or screen) and measures the scores.

2. **I name the type of bias in a sample or a question, and I decide if results can be generalized to a population.**

   Name the type of bias. Explain how it could change the results.

   (a) A TV show asks viewers to go to its website and vote: should the show move to a later time?

   Voluntary response bias. Viewers choose to vote. People with strong feelings vote most, so the result may not match all viewers.

   (b) A survey asks, “Should the city waste money on a new soccer field?”

   Response bias. The word “waste” pushes people to say no.

   (c) A gym emails a survey to 400 randomly chosen members. Only 90 members answer.

   Nonresponse bias. The 90 who answer may be different from the 310 who do not. For example, they may be the happiest or the most upset members.
   Voluntary response: partial credit. The cue is “randomly chosen”, so the answer is nonresponse.

   (d) Lina asks the 25 students in her computer club how many hours a day they use their phones. Can she generalize her results to all students at her school? Explain.

   No. This is a convenience sample, not a random sample. Computer club members may use screens more than other students, so the sample may not represent the school.
   Also accept undercoverage (students not in the club are left out).

5.2 - Sampling Methods

3. **I name and describe simple random, stratified, cluster, systematic, and convenience samples, and I tell stratified and cluster samples apart.**

   A library has 1,500 members. It wants to survey 60 of them about new hours. Name each sampling method.

How the library chooses 60 members	Method
(a) A computer randomly picks 60 member numbers.	Simple random sample
(b) A random member from the first 25, then every 25th member.	Systematic
(c) 20 randomly chosen children, 20 teens, and 20 adults.	Stratified
(d) The first 60 members who walk in on Saturday.	Convenience

(e) A city randomly selects 4 of its 40 apartment buildings. It surveys every person who lives in those 4 buildings. Which sampling method does the city use? Circle the letter.

A Simple random B Stratified Ⓒ Cluster D Systematic

(f) Explain how a stratified sample of the people in the 40 buildings would be different from the city's sample.

A stratified sample takes a few people, chosen at random, from **every** building. The city's cluster sample takes **everyone** from a **few** buildings.

4. **I use a random digit table to choose a sample, and I choose a good sampling method for a situation.**

A coach has 40 players, numbered 01 to 40. Use this row of random digits to choose 4 players.

34603 36904 52041 50165 61689 44176 49590 03997

(a) Write the pairs you read. Then write the 4 players you choose.

Pairs: **34**, 60, **33**, 69, **04**, 52, 04, **15**. Skip 60, 69, 52 (over 40) and the second 04 (a repeat). The sample is players 34, 33, 04, and 15.

A school has 900 students. 300 ride the bus, and 600 do not. The principal wants to ask 90 students about a new bus schedule.

(b) Describe a good sampling method. Name it, and explain why it is a good choice.

Stratified sample. Split the students into bus riders and others. Randomly pick 30 bus riders and 60 others (10% of each). Bus riders may feel differently about the schedule, so both groups are in the sample in the right numbers.
Also accept, with a reason: a stratified sample with equal numbers from each group, a simple random sample, or a systematic sample from the full list of 900.

(c) The principal asks the first 90 students who get off the buses. What is wrong with this method?

It is a convenience sample. Only bus riders are asked. The 600 students who do not ride the bus are left out, so the results are biased.
Also accept undercoverage.

5.3 - Experiments

5. **I name the subjects, treatments, explanatory variable, and response variable in an experiment, and I explain why random assignment matters.**

   A track coach wants to know if a new warm-up helps runners run faster. The 24 runners on the team are randomly split into two groups of 12. One group does the new warm-up. The other group does the usual warm-up. Then each runner runs 400 meters, and the coach records the time.

(a) Subjects	The 24 runners on the team
(b) Treatments	The new warm-up and the usual warm-up
(c) Explanatory variable	The type of warm-up
(d) Response variable	Each runner's time for 400 meters

   (e) Why should the coach split the runners into groups at random? Explain.

   So the two groups are alike. The fastest runners do not all end up in one group. Then a difference in times comes from the warm-up, not from the runners.

   (f) Instead, the coach lets each runner choose a warm-up. Why is this a problem?

   The groups may not be alike. The fastest runners may all choose the new warm-up. Then we cannot tell if the warm-up or the runners caused the faster times.

6. **I explain control groups, placebos, and blinding, and I describe completely randomized, randomized block, and matched pairs designs.**

A company says its new "focus gum" helps students remember. 40 student volunteers are randomly split into two groups of 20. One group chews focus gum. The other chews plain gum that looks and tastes the same. Then all students take the same memory quiz, and a teacher grades it.

(a) Which group is the control group?

The 20 students who chew the plain gum

(b) The plain gum is a placebo. Why give the control group a placebo, and not nothing at all?

Students may do better just from *thinking* they got focus gum (the placebo effect). With a placebo, both groups think so. Then an extra difference comes from the gum itself.

(c) The teacher who grades the quizzes does not know which gum each student chewed. Why?

If she knew, she might, without meaning to, grade the focus gum group more kindly. Not knowing keeps the grading fair.

(d) This experiment is a completely randomized design. Describe how to test the gum with a matched pairs design instead.

Each student takes two memory quizzes of the same difficulty: once with focus gum and once with plain gum. A coin flip decides which gum comes first. Compare each student's two scores. Also accept: pair students with similar scores (for example, on a pretest), then flip a coin in each pair to choose who gets focus gum.

(e) 16 of the 40 students are in 9th grade and 24 are in 12th grade. Older students may remember more. Describe how to test the gum with a randomized block design.

Make two blocks: the 9th graders and the 12th graders. In each block, randomly assign half to focus gum and half to plain gum (8 and 8 of the 9th graders, 12 and 12 of the 12th graders). Compare the two gums inside each block.

5.4 - What a Study Can Prove

7. **I decide if a study's results can be generalized to a population, and I decide if a study can show cause and effect.**

   For each study, can we conclude cause? Can we generalize to the population? Explain.

Study	Cause? Why?	Generalize? Why?
(a) Researchers randomly choose 600 teens in Missouri. Teens who spend more time outside sleep more.	No. Observational study, no random assignment.	Yes, to Missouri teens. The sample is random.
(b) 60 volunteers from a typing class are randomly assigned to Game A or Game B. The Game A group types faster.	Yes, for these 60 (random assignment). Also accept: yes, if the difference is statistically significant.	No. Volunteers are not a random sample.

(c) A researcher randomly selects 200 students from a school district. She randomly assigns half to study with flash cards and half with a video. The flash card group scores higher, and the difference is statistically significant. Which conclusion is best supported?

A Studying with flash cards causes higher scores for all students in the United States.

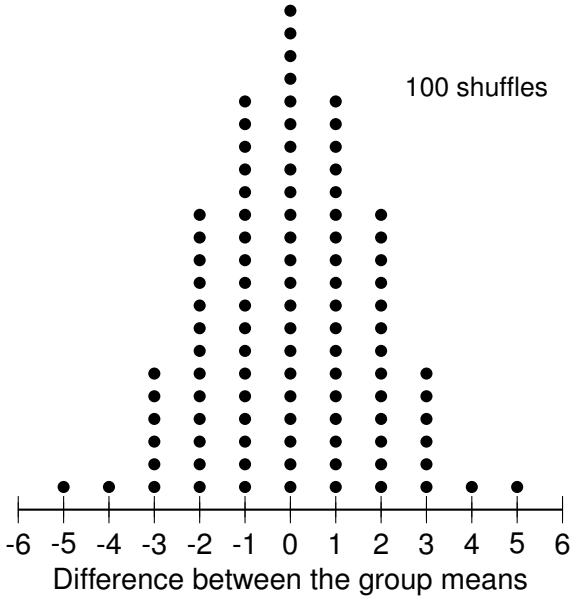
Ⓑ Studying with flash cards causes higher scores for students in this school district.

C Flash cards are associated with higher scores, but the study cannot show cause.

D No conclusion can be made, because not every student in the district was studied.

8. **I use a simulation to decide if a result is statistically significant, and I find mistakes in conclusions about studies.**

A teacher tests a new vocabulary game. 30 students are randomly split into two groups of 15. The game group's mean is 4 points higher. Each dot is the difference between the group means in one random shuffle of the 30 scores, when the game does nothing.



(a) How many shuffles gave a difference of 4 or more?

1 + 1 = 2 shuffles

(b) Is a 4-point difference statistically significant? Explain.

Yes. Chance gave a difference this big only 2 of 100 times (2%). That is less than 5%. The game probably helped.

(c) Find the mistake. Suppose the game group had scored only 2 points higher. The teacher still says, "The game works."

Chance gave 2 or more 13 + 6 + 1 + 1 = 21 times out of 100 (21%). That is common. 2 points is not statistically significant. It could be chance.

(d) Find the mistake. A survey of 500 randomly chosen adults finds that dog owners walk more. A headline says, "Owning a dog makes people walk more."

A survey shows an association, not a cause. People who already like to walk may choose to own a dog.